King County Residential Real Estate Analysis

Building a High-Precision Machine Learning Pipeline for Market Prediction

Using Gradient Boosting to identify value drivers in a complex, multi-segmented market spanning 21,000+ transactions.

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The Dataset: Scope & Complexity

Context: King County (Seattle area) transactions from May 2014 to May 2015.

Data Volume: 21,613 observations across 21 structural and geographic features.

Target: Sale Price (Median: $450,000 | Max: $7.7M).

Key Challenge: The market is heavily right-skewed; predicting the standard market is different from predicting the top 1% (Luxury segment).

The Mission: Build a robust ML pipeline to predict sales prices.
Identify the "Golden Features" that dictate property value.
Stress-test the model against the Luxury Segment (≥$650k).

Columns Description: This dataset comprises one-year data (from May 2014 to May 2015) of house sale prices across King County, including Seattle. It features 21 different columns, providing a comprehensive overview of the real estate market:

- id: A unique identifier for a house.
- date: The date on which the house was sold.
- price: The sale price of the house (prediction target).
- bedrooms: Number of bedrooms in the house.
- bathrooms: Number of bathrooms in the house, per bedroom.
- sqft_living: Square footage of the interior living space.
- sqft_lot: Square footage of the land space.
- floors: Number of floors (levels) in the house.
- waterfront: Whether the house has a waterfront view.
- view: Number of times the house has been viewed.
- condition: The overall condition of the house.
- grade: The overall grade given to the house, based on the King County grading system.
- sqft_above: Square footage of the house apart from the basement.
- sqft_basement: Square footage of the basement.
- yr_built: The year the house was built.
- yr_renovated: The year the house was renovated.
- zipcode: ZIP code area.
- lat: Latitude coordinate.
- long: Longitude coordinate.
- sqft_living15: The interior living space for the nearest 15 neighbors in 2015.
- sqft_lot15: The land spaces for the nearest 15 neighbors in 2015.
- **TARGET --> Price**: Our primary focus is to understand which features most significantly impact the house price. Additionally, we aim to explore properties valued at $650K and above for more detailed insights.

Data Integrity: Logic-Driven Cleaning

The Problem with "Black-Box" Cleaning: Using standard statistical filters (IQR) often deletes high-value luxury data.

Our Solution: Domain Logic.

Identified Typos: Removed a "33-bedroom" house with only 1,600 sqft (confirmed data entry error).
Invalid Entries: Filtered properties with 0 bedrooms/bathrooms (likely land-only or errors).
Validated Zeros: Preserved "0 sqft basement" and "0 yr_renovated" as valid market states, not missing values.

STATS SUMMARY

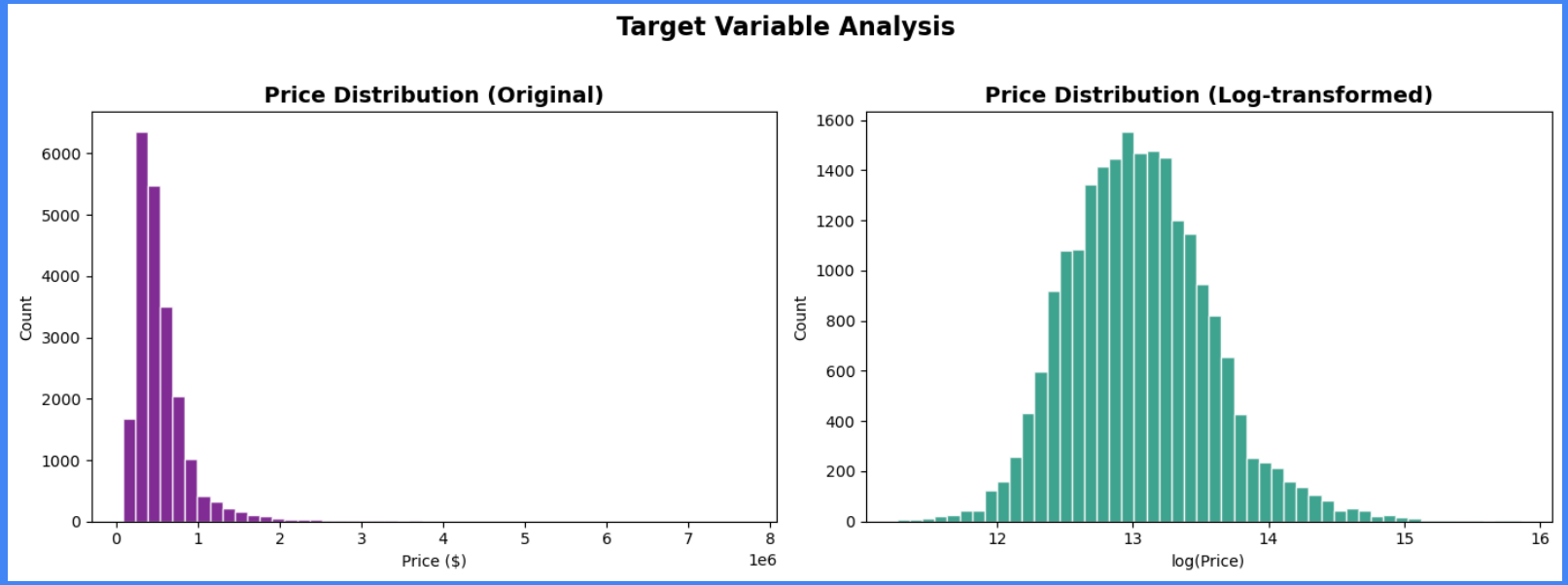
	id	price	bedrooms	bathrooms	sqft_living	sqft_lot	floors	waterfront	view	condition	grade
count	2.161300e+04	2.161300e+04	21613.000000	21613.000000	21613.000000	2.161300e+04	21613.000000	21613.000000	21613.000000	21613.000000	21613.000000
mean	4.580302e+09	5.400881e+05	3.370842	2.114757	2079.899736	1.510697e+04	1.494309	0.007542	0.234303	3.409430	7.656873
std	2.876566e+09	3.671272e+05	0.930062	0.770163	918.440897	4.142051e+04	0.539989	0.086517	0.766318	0.650743	1.175459
min	1.000102e+06	7.500000e+04	0.000000	0.000000	290.000000	5.200000e+02	1.000000	0.000000	0.000000	1.000000	1.000000
25%	2.123049e+09	3.219500e+05	3.000000	1.750000	1427.000000	5.040000e+03	1.000000	0.000000	0.000000	3.000000	7.000000
50%	3.904930e+09	4.500000e+05	3.000000	2.250000	1910.000000	7.618000e+03	1.500000	0.000000	0.000000	3.000000	7.000000
75%	7.308900e+09	6.450000e+05	4.000000	2.500000	2550.000000	1.068800e+04	2.000000	0.000000	0.000000	4.000000	8.000000
max	9.900000e+09	7.700000e+06	33.000000	8.000000	13540.000000	1.651359e+06	3.500000	1.000000	4.000000	5.000000	13.000000

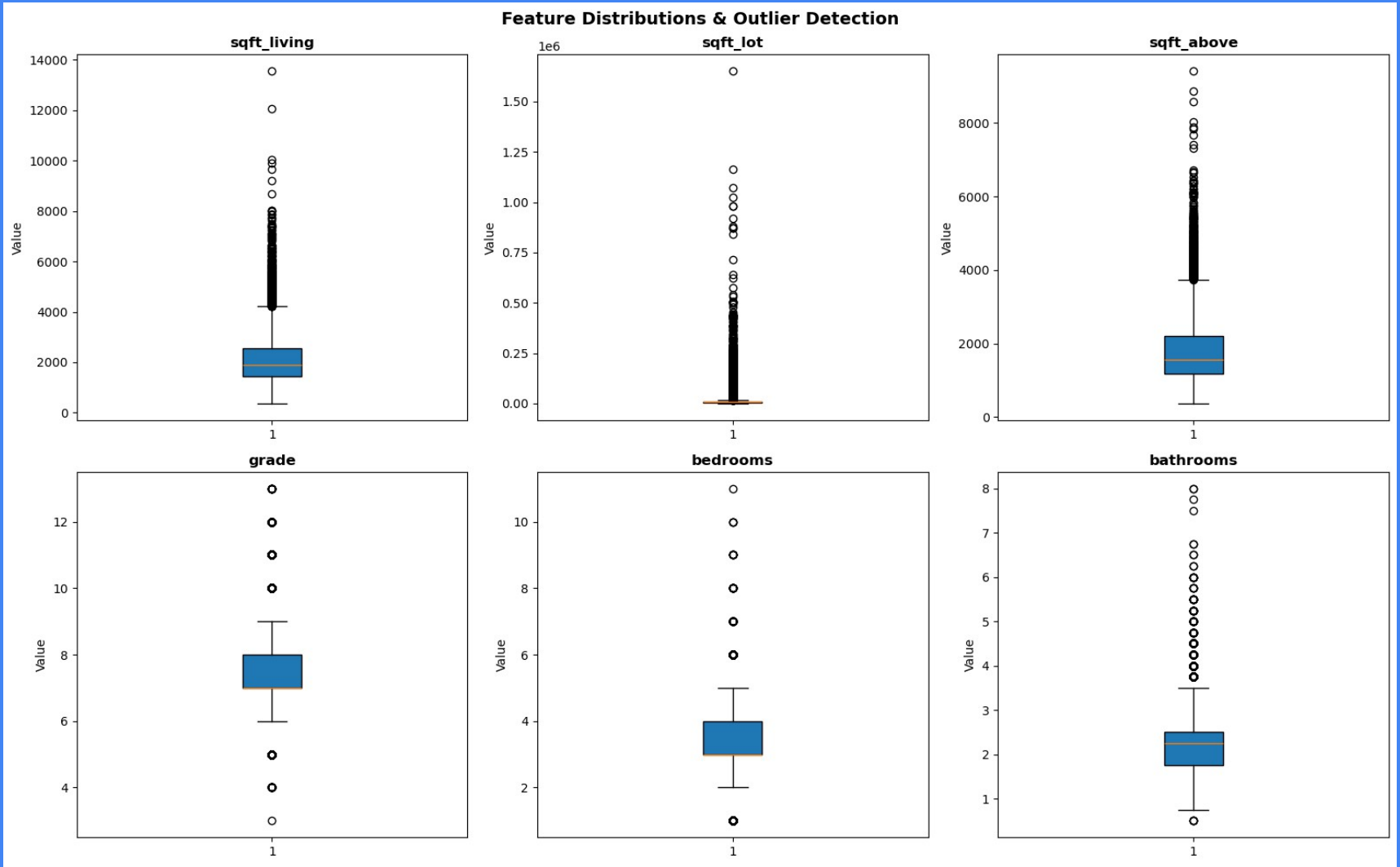
Target Transformation: Correcting Market Skew

The Challenge: Real estate prices follow a power-law distribution.
Skewness: 4.026 (Highly problematic for Linear Models).

The Strategy: Applied Log-Transformation (log(price + 1)).Resulting Skewness: 0.430 (Near-normal distribution).

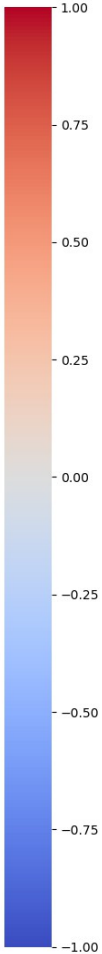
Why it matters: This normalization stabilizes variance and significantly reduces the impact of extreme outliers on the model's error rate.




Feature Distributions & Outlier Detection
sqft_living
sqft_lot
sqft_above
grade
bedrooms
bathrooms
Value
1e6
1

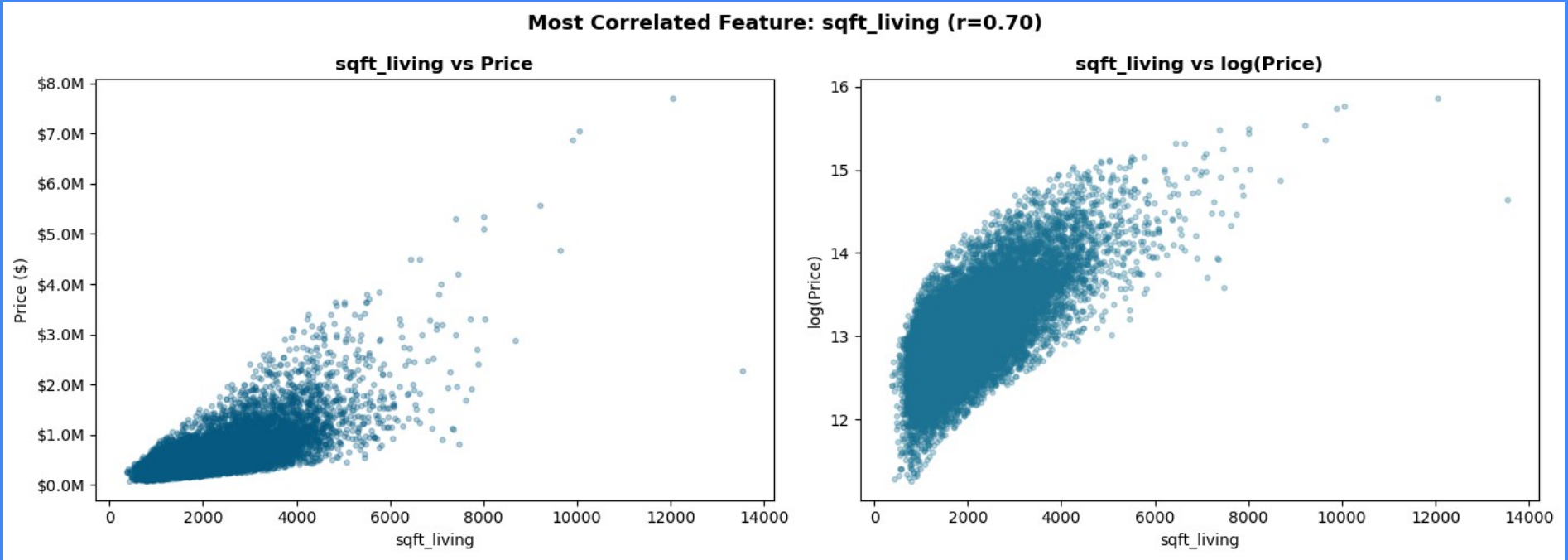
Correlation Matrix — King County Houses

	id	price	bedrooms	bathrooms	sqft_living	sqft_lot	floors	waterfront	view	condition	grade	sqft_above	sqft_basement	yr_built	yr_renovated	zipcode	lat	long	sqft_living15	sqft_lot15
id																				
price	-0.02																			
bedrooms	0.00	0.32																		
bathrooms	0.01	0.53	0.53																	
sqft_living	-0.01	0.70	0.59	0.76																
sqft_lot	-0.13	0.09	0.03	0.09	0.17															
floors	0.02	0.26	0.18	0.50	0.35	-0.00														
waterfront	-0.00	0.27	-0.01	0.06	0.10	0.02	0.02													
view	0.01	0.40	0.08	0.19	0.28	0.07	0.03	0.40												
condition	-0.02	0.04	0.02	-0.13	-0.06	-0.01	-0.26	0.02	0.05											
grade	0.01	0.67	0.37	0.67	0.76	0.11	0.46	0.08	0.25	-0.15										
sqft_above	-0.01	0.61	0.49	0.69	0.88	0.18	0.52	0.07	0.17	-0.16	0.76									
sqft_basement	-0.01	0.32	0.31	0.28	0.44	0.02	-0.25	0.08	0.28	0.17	0.17	-0.05								
yr_built	0.02	0.05	0.16	0.51	0.32	0.05	0.49	-0.03	-0.05	-0.36	0.45	0.42	-0.13							
yr_renovated	-0.02	0.13	0.02	0.05	0.06	0.01	0.01	0.09	0.10	-0.06	0.01	0.02	0.07	-0.22						
zipcode	-0.01	-0.05	-0.16	-0.20	-0.20	-0.13	-0.06	0.03	0.08	0.00	-0.19	-0.26	0.07	-0.35	0.06					
lat	-0.00	0.31	-0.01	0.02	0.05	-0.09	0.05	-0.01	0.01	-0.02	0.11	-0.00	0.11	-0.15	0.03	0.27				
long	0.02	0.02	0.14	0.22	0.24	0.23	0.13	-0.04	-0.08	-0.11	0.20	0.34	-0.14	0.41	-0.07	-0.56	-0.14			
sqft_living15	-0.00	0.59	0.40	0.57	0.76	0.14	0.28	0.09	0.28	-0.09	0.71	0.73	0.20	0.33	-0.00	-0.28	0.05	0.34		
sqft_lot15	-0.14	0.08	0.03	0.09	0.18	0.72	-0.01	0.03	0.07	-0.00	0.12	0.20	0.02	0.07	0.01	-0.15	-0.09	0.26	0.18	



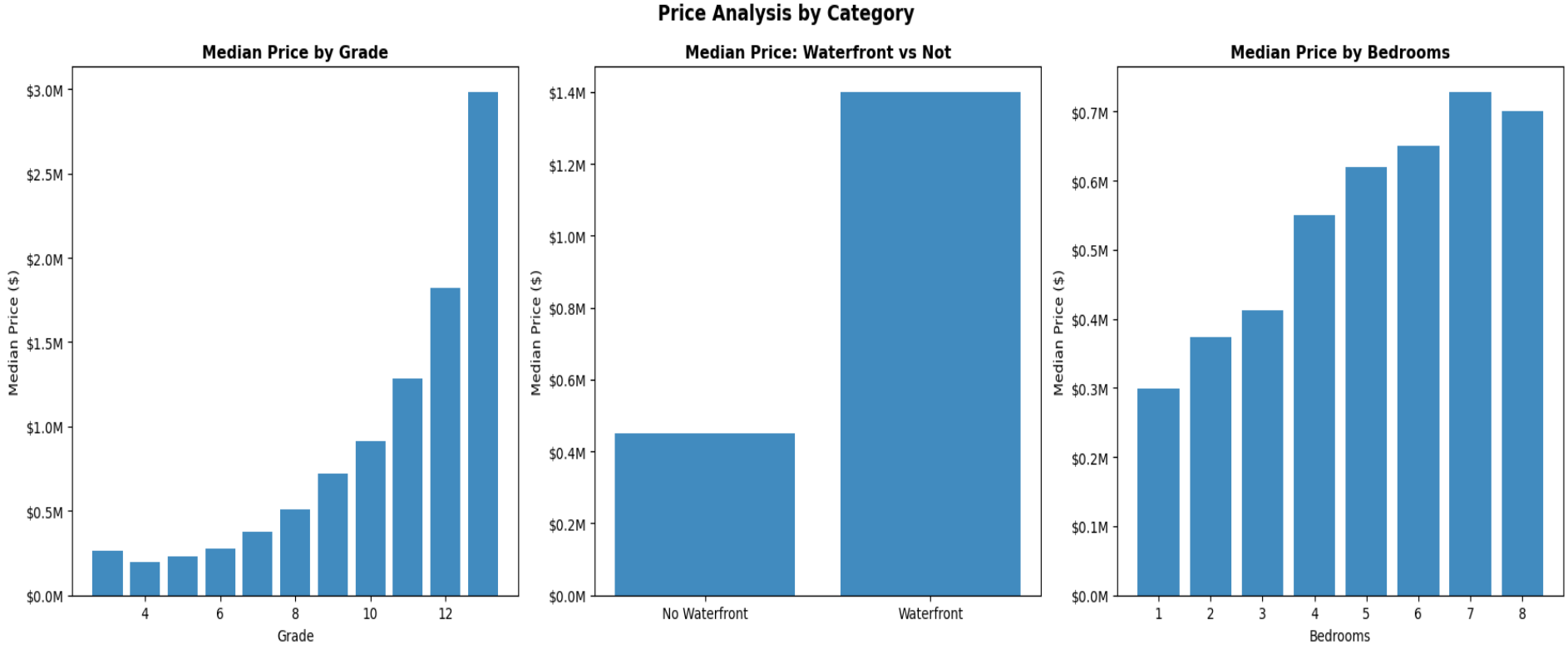
EDA: Identifying the Value Drivers

The "Product" vs. The "Foundation": Strongest linear correlation: sqft_living (r=0.70).



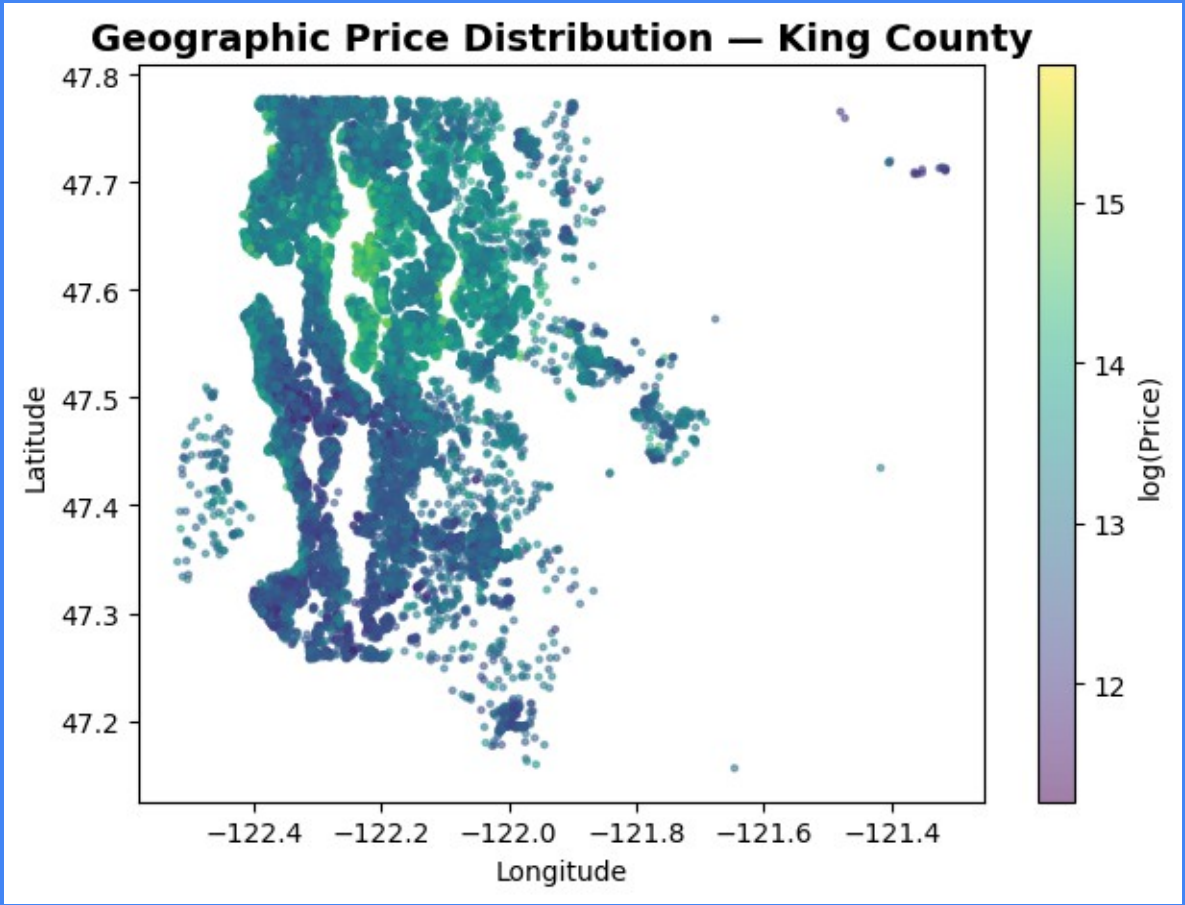
The "Waterfront Premium":

Median prices for waterfront properties are 3x higher than inland homes.



Geographic Clustering:

Higher prices cluster around central Seattle and specific coastal zipcodes.



Feature Engineering: Adding Intelligence

Moving from Raw Data to Insights:

House Age: Converted "Year Built" into a meaningful "Age" metric.

Renovation Recency: Calculated years since the last upgrade to capture asset appreciation.

Living vs. Neighbors: A ratio of the home's size relative to the closest 15 neighbors—capturing whether a house is the "best in the block."

Lot Utilization: House-to-land ratio to differentiate urban density from suburban estates.

Model Selection: The Competitive Edge

Objective: Find the model that captures non-linear relationships (e.g., how the value of an extra bedroom decreases as the house gets older).

Performance Ranking (R^2):

Gradient Boosting: 0.7408 (Winner)
Random Forest: 0.7268
Lasso/Ridge: 0.7048

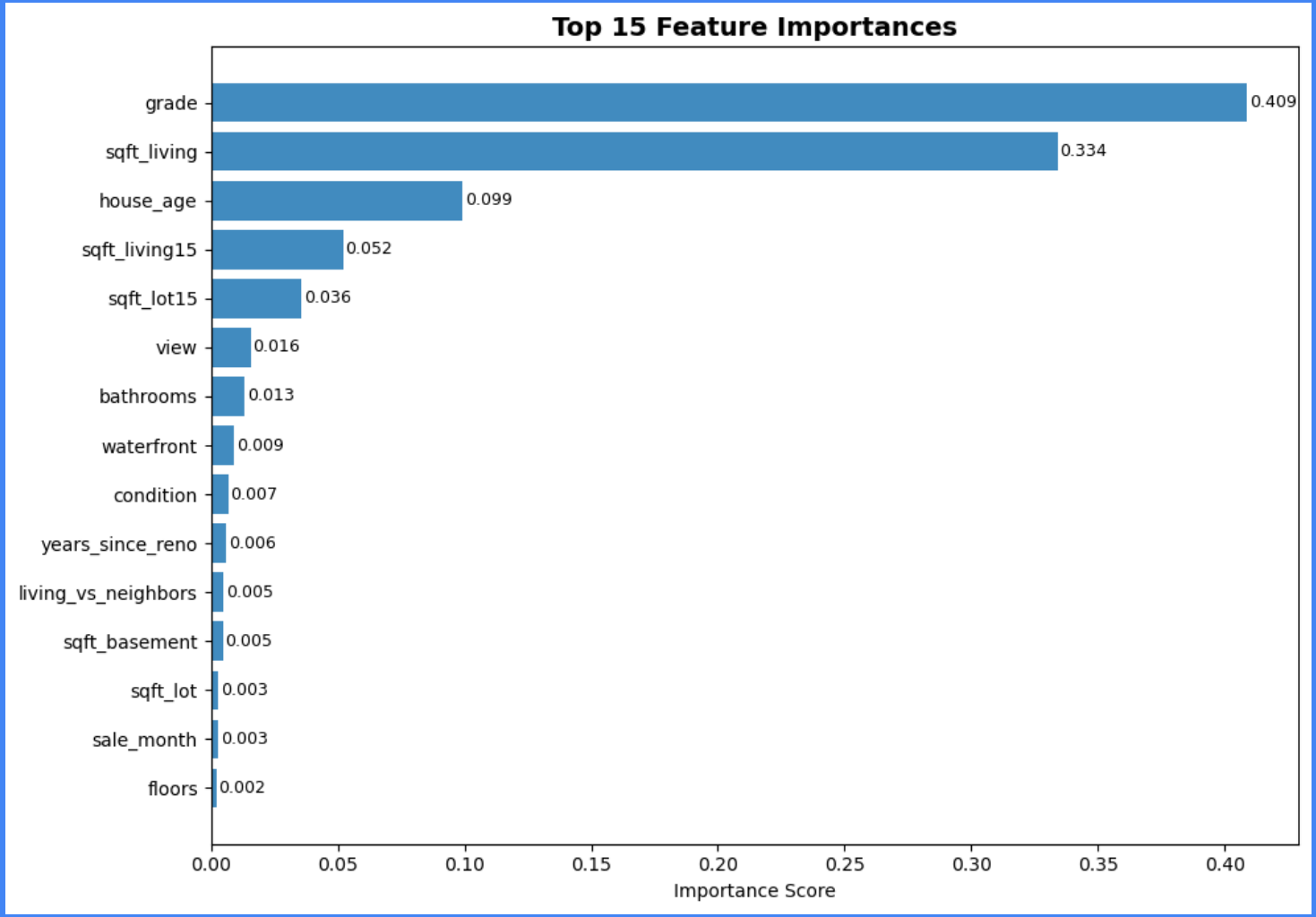
The Technical Verdict: Ensemble tree models (GBM) outperformed linear regressions because they effectively model interactions between features like grade and house_age.

What Moves the Needle? (Feature Importance)

Surprising Insight: Grade (40.9% importance) outranks sqft_living (33.4%).

Interpretation: In King County, quality of construction and finishes (Grade) is a stronger predictor of value than raw interior square footage.

Secondary Factors: House age and the neighborhood context (sqft_living15) provide the remaining predictive power.


Top 15 Feature Importances
grade
sqft_living
house_age
sqft_living15
sqft_lot15
view
bathrooms
waterfront
condition
years_since_reno
living_vs_neighbors
sqft_basement
sqft_lot
sale_month
floors
0.409
0.334
0.099
0.052
0.036
0.016
0.013
0.009
0.007
0.006
0.005
0.005
0.003
0.003
0.002
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
Importance Score

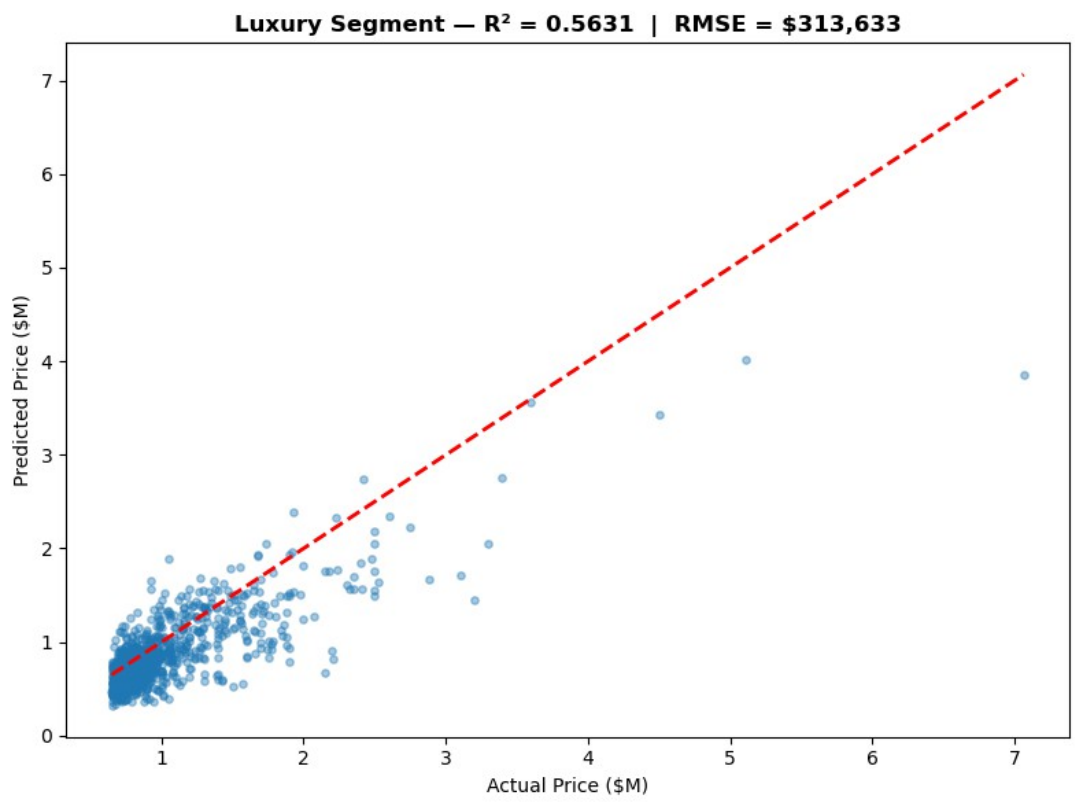
The "Luxury Ceiling" Analysis

Defining Luxury: Properties > $650,000 (24.6% of the market).

The Predictive Gap: General Market R^2: 0.74

Luxury Segment R^2: 0.56 (RMSE jumps to $313K).

Finding: High-end homes are "idiosyncratic." Their value depends on factors not captured in standard data (architectural pedigree, specific views, custom finishes).



Conclusion & Strategic Roadmap

Current State: We have a robust model that automates valuation for 75% of the market with high confidence.

Future Roadmap:

Specialized Modeling: Develop a dedicated regressor for the Luxury segment using hyper-local features.

External Data: Integrate School District ratings and proximity to major employers (Amazon/Microsoft/Starbucks).

Hyperparameter Optimization: Use RandomizedSearchCV to fine-tune the Gradient Boosting learning rate.